

Is GenAI Worth It for Ad Creatives? A Small Business Case with a Rating Survey

Shady Shiha · October 2026 · Code: <https://github.com/shdyhssein/StuFoodproj.git>

1. Business question and summary

A marketing team wants to test several ad variants before spending media budget. Producing them by hand is slow, so the question is: **at what volume does a GenAI pipeline beat manual production on cost, and are the results good enough to use?**

I built a small Python pipeline that generates ad copy and image prompts for a fictional student meal-prep app, logs cost, tokens and latency per asset, and tested 7 ads in a rating survey (n = 14). Main findings:

- **Cost:** Under my assumptions, GenAI costs about €5 per usable asset versus €25-100 manually (a 79-95% saving). The model breaks even after roughly 3-13 usable assets, depending on how long manual production takes. The manual times are assumptions, not measurements.
- **Quality:** Ratings differed strongly between ads. Five ads were rated well, two clearly poorly. The gap between a good and a poor ad of the *same* creative angle was far larger than any gap between angles.
- **Implication:** GenAI makes producing many candidates cheap, but a human selection step and a quick pre-test are what separate usable from unusable ads.

2. Method

Pipeline. A config file holds the creative brief (product, audience, tone, key message) and five creative angles (rational, emotional, humorous, social, driven). A Python script calls a Gemini text model (gemini-3.1-flash-lite) and returns a headline, a short body text and an image prompt per asset. Every asset is logged to a CSV: token counts, cost, latency, retries, and an `ai_label` flag. I generated 30 candidate assets. Image prompts were turned into images for 7 selected assets using a chat-based image tool, which was a manual step. The image cost is therefore modeled at list price (about €0.03 per image), not measured.

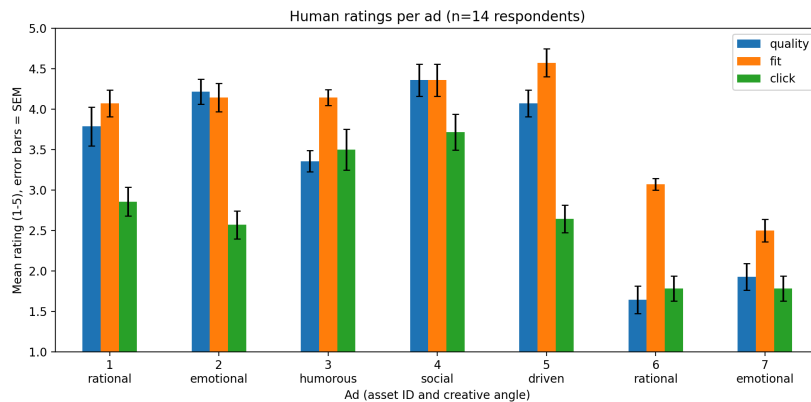
Survey. 14 respondents (7 per ad order, with the second group seeing the ads in reverse order) rated each of the 7 ads on three 1-5 scales (overall quality, fit for students, click intent) and answered whether the ad looks AI-made. Recruitment was through friends and university groups (convenience sample).

Cost model. Cost per *usable* asset for manual versus GenAI production, plus a one-off setup cost, a break-even volume, scenarios and a Monte Carlo simulation. Measured inputs: API cost per attempt (about €0.029, including the modeled image price), image time (1.6 min), review time (1.0 min), and the usable rate (71%: 5 of 7 ads had a mean quality of at least 3). Assumed inputs: manual time per asset (30 / 60 / 120 min), 2 min of prompt and selection time per attempt, 5 h of setup, and hourly rates of €15 / €50 / €100.

3. Results

3.1 Survey ratings (mean, scale 1-5)

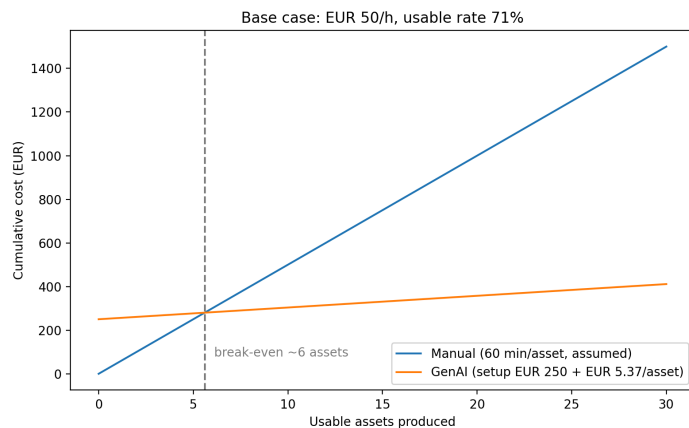
Ad	Angle	Quality	Fit	Click intent
1	rational	3.79	4.07	2.86
2	emotional	4.21	4.14	2.57
3	humorous	3.36	4.14	3.50
4	social	4.36	4.36	3.71
5	driven	4.07	4.57	2.64
6	rational	1.64	3.07	1.79
7	emotional	1.93	2.50	1.79



- The ads differ clearly (Friedman test, $p < 0.001$ for all three measures; Kendall's $W = 0.71$ for quality, 0.62 for fit and 0.68 for click intent). This is driven mostly by ads 6 and 7. Without them, click intent still differs ($W = 0.63$, $p < 0.001$), quality differs more weakly ($W = 0.26$, $p = 0.006$) and fit does not ($p = 0.17$).
- **Execution mattered more than the angle.** The rational angle includes ad 1 (quality 3.79) and ad 6 (1.64). The emotional angle includes ad 2 (4.21) and ad 7 (1.93). With one or two ads per angle, no conclusion about the angles themselves is possible.
- **Ad 3 (humorous) has the second-highest click intent (3.50) but the lowest quality of the top five (3.36).** It also uses a cartoon style unlike the others, so humor and visual style cannot be separated.
- No order effect was visible: the two weakest ads were rated low in both orders, and the two versions did not differ significantly (all $p \geq 0.13$).
- Ads 1-5 were judged AI-made by all respondents, while ads 6 and 7 got some "not sure" answers (36% and 43%). Detectability is at its ceiling, and the question may have primed respondents.

3.2 Cost model (usable rate 71%, €50/h)

Manual time per asset	Manual cost	GenAI cost	Saving	Break-even
30 min	€25.00	€5.37	79%	about 13 usable assets
60 min	€50.00	€5.37	89%	about 6 usable assets
120 min	€100.00	€5.37	95%	about 3 usable assets



- In a Monte Carlo simulation of 10,000 draws (manual time 30-120 min, usable rate 40-100%, prompt time 1-5 min), GenAI was cheaper in every draw, with a median saving of 88% (5th-95th percentile: 75-94%).
- **Tipping point:** in the base case, GenAI would still be cheaper until each attempt takes about 43 human minutes (about 21 minutes against a 30-minute manual asset). My estimate of 4.6 minutes is surely too low, because it leaves out brief writing and the 23 candidates that never became images, but the conclusion has a wide margin.
- The hourly rate changes the euro amounts but barely the percentages, because manual cost, GenAI cost and setup cost all scale with it. It is therefore not a real sensitivity dimension. The heatmap (manual minutes versus human minutes per GenAI attempt) shows the drivers that do matter.
- A sensitivity heatmap of the break-even volume is in docs/breakeven_heatmap.png, and the ads tested are in docs/ads/.

4. Governance and risks

- Transparency: AI-generated content may need to be labeled under EU AI Act transparency rules. The exact scope and timing should be checked against the current legal text. This is not legal advice. The pipeline marks every asset with `ai_label = true`, so labeling can be automated downstream.

- **Brand safety and bias:** generated images can reproduce stereotypes or show inappropriate content. The prompt forbids real people's likenesses, and every ad is reviewed by a human before use. Image prompts that showed groups of people were avoided for the survey.
- **Rights and traceability:** generated images raise open copyright questions. Logging model, prompt version and timestamp per asset gives an audit trail.
- **Human in the loop:** selection and review stay with a person, and the survey shows why: 2 of 7 ads were rated poorly.

5. Limitations

- **Small convenience sample** (n = 14, mostly engineering, business and economics students) and exploratory tests only. Angles are unbalanced, with one or two ads each.
- **No measured manual baseline.** Manual production times are assumptions, so the savings are scenario results and not observed facts.
- **Usable rate** is a proxy from 7 ads that were selected from 30 candidates. Image creation was done in a chat tool, so image time covers only that step, and image cost is modeled.
- **The product name was not enforced in the prompt.** Several ad texts use "StuFood" and others do not, which may influence ratings. Ad 3 differs in image style.
- Ratings use single items, and the "looks AI-made" question may have primed respondents.

6. Recommendation and next steps

For a team producing creatives regularly, the model supports using GenAI to generate many candidates cheaply, followed by human selection and a short panel pre-test before media spend. Next steps: measure a real manual baseline, automate the image step through an API, add an LLM-as-judge pre-screen and compare it with the human ratings, and run a field test with real click data.